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PyStarshade: simulating high-contrast imaging of exoplanets with starshades

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Software

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Summary

Starshades are external occulting masks designed to suppress starlight by a factor of 10^{-11} or more to image faint exoplanets. Beyond this overarching task lie specific questions about the properties and configurations of exoplanetary systems that can be imaged, which spectral features can be characterized, and which instrument designs are most suitable. `PyStarshade` is a Python toolbox to efficiently perform optical simulations of starshades with exoplanetary scenes to assess the utility of starshades in future direct-imaging missions.

Statement of need

`PyStarshade` provides a toolbox for performing optical simulations from source to focal plane with a configurable starshade telescope design. Complex electric fields are calculated at three planes of propagation (starshade, telescope aperture and focal plane) using Fresnel or Fraunhofer diffraction formula where appropriate. First-order imaging characteristics of a starshade can be determined from analytic relations which depend on the starshades size, telescope aperture size, wavelength and flight distance. `PyStarshade` allows for second order imaging characteristics to be studied, including simulations of imaging for a discretized exoplanetary scene of flux, varying starshade mask and telescope aperture mask to study throughput and post-processing methods. An example is provided on using spectral- and time-dependent scenes consisting of planets, star and dust-disk, generated with `ExoVista` (Stark, 2022), the output imaging is shown in Figure 1. `PyStarshade` is intended to be flexible and efficient in studying i) exoplanet retrievals, and, ii) instrument design.

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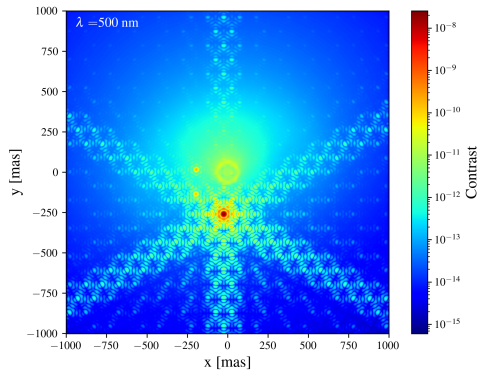

Figure 1: Simulated imaging of a synthetic exoscene (ExoVista) with three visible exoplanets at a wavelength of 500 nm. A 60 m starshade configuration and a 6m segmented pupil was used for this example.

Aimed at being broadly useful for numerically intense Fourier optical simulations, tools are provided for efficient Fresnel and Fraunhofer propagation using Fourier spectral sampling with Bluestein Fast Fourier Transforms (FFT). Furthermore, a novel technique to chunk these FFT calculations is implemented and mitigates memory bottlenecks. A set of unit tests are provided to validate the Fourier diffraction tools. 'PyStarshade' pre-computes fields, point-spread-functions and throughput grids for different stages of the optical propagation chain, allowing for flexibility in modifying instrument parameters or telescope aperture masks. 'PyStarshade' optionally interfaces with HCIPy (Por et al., 2018) to generate telescope apertures and the resulting imaging in a streamlined fashion.

Related software

The Starshade Imaging Simulation Toolkit for Exoplanet Reconnaissance (`SISTER`) (Hildebrandt et al., 2022) implemented in Matlab performs detailed end-to-end simulations of starshade imaging, further including instrument noise sources such as solar glint, and tools for modeling an exoplanetary system and background objects. The `SISTER` tool performs diffraction calculations from starshade to telescope aperture using the boundary diffraction wave method due to (Cady, 2012). `Fresnaq` (Barnett, 2020b) is a Matlab code to efficiently compute diffraction of a starshade at the telescope aperture via the non-uniform FFT (Barnett, 2020a). `Diffraq` (Harness, 2021) is a Python implementation of `Fresnaq`, this toolbox also provides an implementation of the boundary diffraction wave method and tools to generate starshade masks (Harness et al., 2018). The Exoplanet Open-Source Imaging Mission Simulator (`EXOSIMS`) (Savransky et al., 2017) provides methods for orbital simulation of starshades and fuel consumption calculations.

Acknowledgements

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